DVĪPAHAṂSA

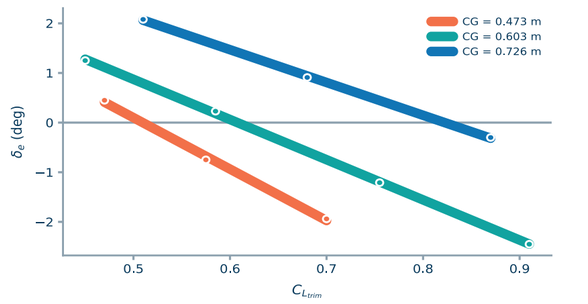
ISLAND HOPPER AMPHIBIAN AIRCRAFT

Aavash Dhakal, Ananya Rijal, Mohan Prakash Yadav, Roshan Giri

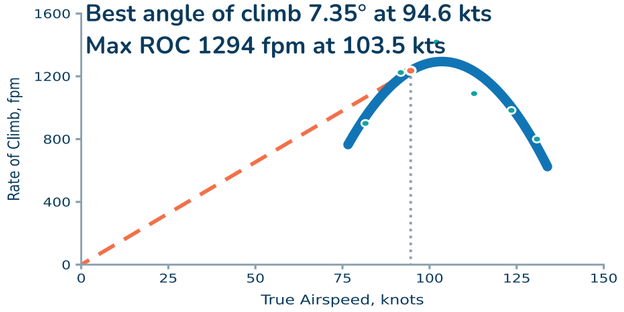
IOE Pulchowk Campus



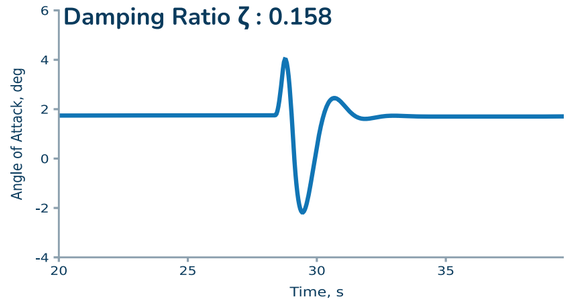
Elevator Trim Vs Cl



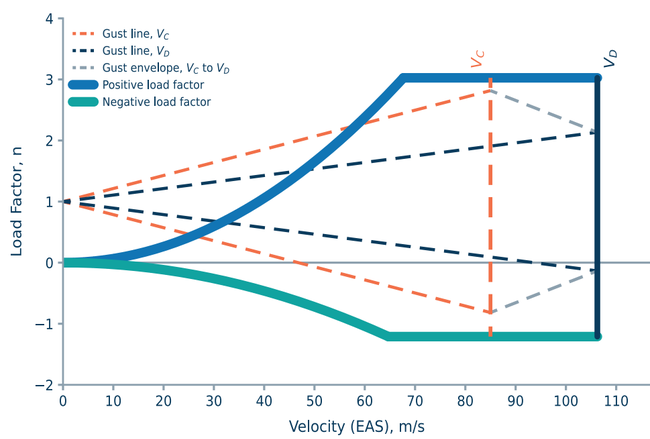
Rate Of Climb (ROC)



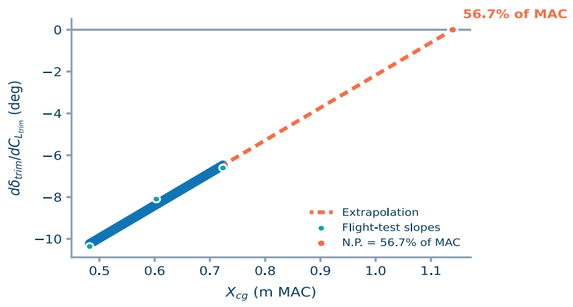
Short Period



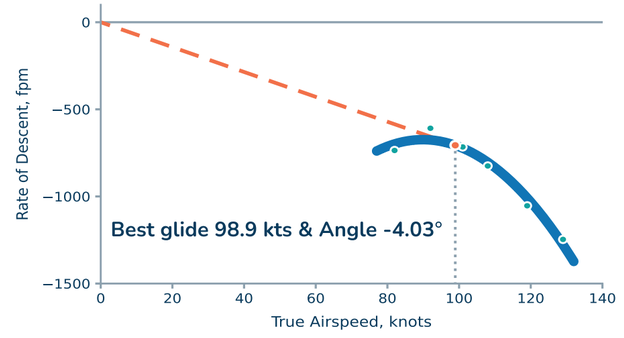
V-n Diagram



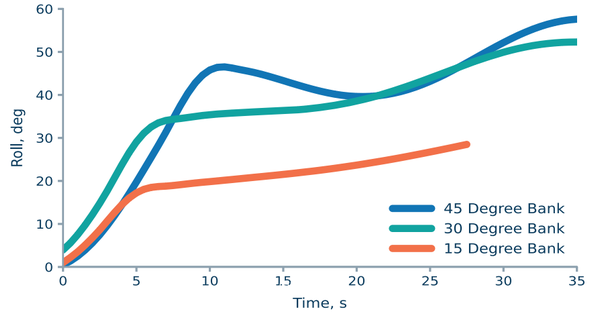
Determination Of Neutral Point



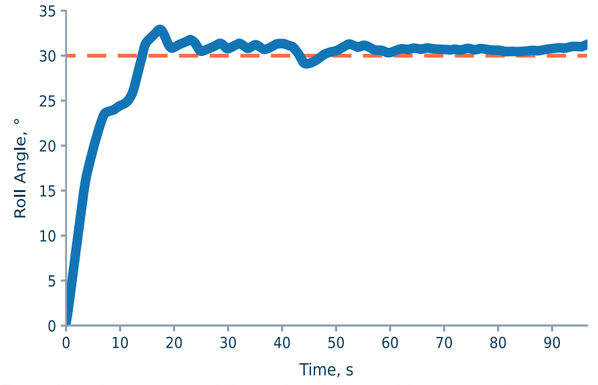
Rate Of Descent (ROD)



Spiral Mode

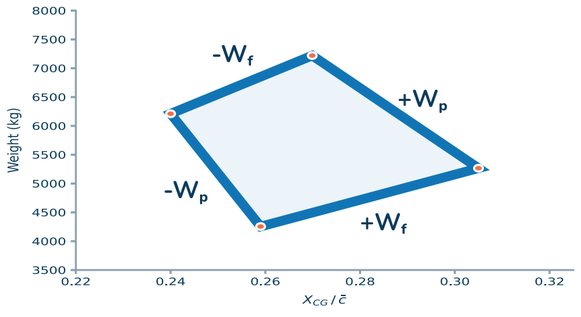


Handling Quality Test

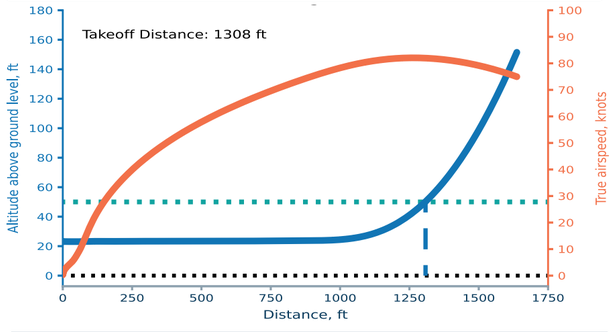


The aircraft was rated 3 on the Cooper-Harper scale, which falls in the "Fair" category and sits at the upper end of Level 1 handling qualities. A rating of 3 means the aircraft had some mildly unpleasant deficiencies, but these were minor and did not prevent the pilot from reaching the desired performance. Only minimal pilot compensation was required.

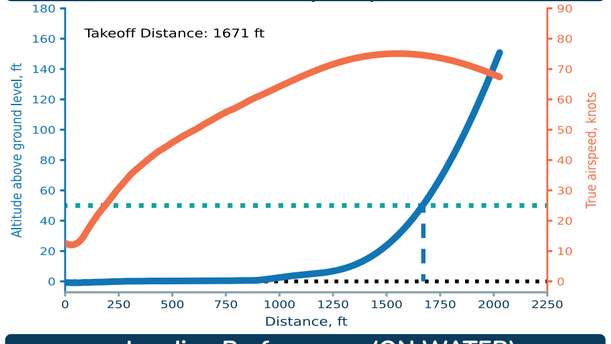
CG Travel Diagram



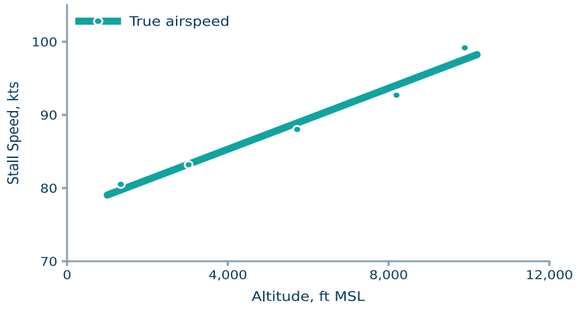
Take-Off Performance (ON GROUND)



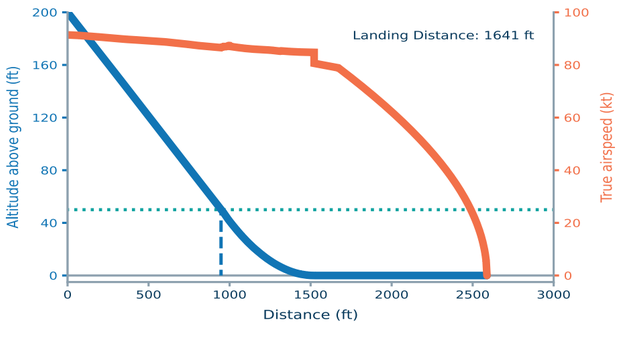
Take-Off Performance (ON WATER)



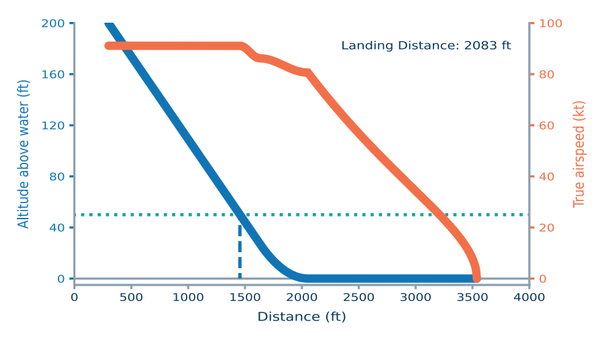
Stall Characteristics



Landing Performance (ON GROUND)



Landing Performance (ON WATER)



Conclusion

Dvipahamsa successfully completed its preliminary flight tests, confirming the design's suitability for island-hopping operations from both water and land. The aircraft was rated 3 on the Cooper-Harper scale, which means only mild deficiencies and minimal pilot compensation. It showed a well-damped dutch roll response and a positive static margin across its full CG envelope. It cruises at 85 m/s at 9,000 ft and covers a design range of 1550 km. This lets it carry 17 passengers and 3 crew between remote atolls and other water-bound locations that lack runway infrastructure.